

Hilton Stammbach 2

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1 Categories

Claim: Let $f : A \rightarrow B, g : B \rightarrow C$ be isomorphisms. Then $gf : A \rightarrow C$ is an isomorphism.

Proof: Pick $f' : B \rightarrow A$ and $g' : C \rightarrow B$ such that $f'f = 1_A, ff' = 1_B, g'g = 1_B, gg' = 1_C$. Then $f'g'gf = f'1_Bf = f'f = 1_A$ and $gff'g' = g1_Bg' = gg' = 1_C$.

A category $\mathcal{C}$ is said to be a zero category if there exists $0 \in \mathcal{C}$ such that $\text{Hom}(0, B), \text{Hom}(A, 0)$ both have one element for all $A, B \in \mathcal{C}$. I claim that the zero object 0 is determined up to isomorphism.

Proof: Suppose $0, 0'$ are zero objects. $\text{Hom}(0, 0), \text{Hom}(0, 0'), \text{Hom}(0', 0), \text{Hom}(0', 0')$ all have one element, say $1_0, f, g, 1_{0'}$. Then $gf \in \text{Hom}(0, 0)$, so $gf = 1_0$. Similarly, $fg = 1_{0'}$, so $0 \cong 0'$.

Claim: Suppose $A, B \in \mathcal{C}$. There exists a unique map $0_{AB} \in \text{Hom}(A, B)$ such that there exists $0 \in \mathcal{C}$ such that 0_{AB} factors as $A \rightarrow 0 \rightarrow B$.

Proof: Suppose $0, 0'$ are zero elements. Pick $f \in \text{Hom}(A, 0), g \in \text{Hom}(0, B), f' \in \text{Hom}(A, 0'), g' \in \text{Hom}(0', B)$. We wish to show $gf = g'f'$. Pick $\psi \in \text{Hom}(0, 0')$ to be the isomorphism. Then $\psi f \in \text{Hom}(A, 0')$, so $\psi f = f'$. Let ψ' be the inverse of ψ . Then $g\psi' \in \text{Hom}(0', B)$, so $g\psi' = g'$. Thus, $g'f' = g\psi'\psi f = g1_0f = gf$.

Claim: Let $f : A \rightarrow B$ be an isomorphism. Then the inverse g is unique.

Proof: Suppose g, g' are inverses. Then $g = g1_B = gfg' = 1_Ag' = g'$. Denote the inverse by f^{-1} .

Claim: $(f^{-1})^{-1} = f$.

Proof: We have $f^{-1}f = 1_A$ and $ff^{-1} = 1_B$, so treating f^{-1} as the base and f as the inverse, we have $f = (f^{-1})^{-1}$.

Claim: Let $f : W \rightarrow X$, $g : Y \rightarrow Z$, then $0_{XY}f = 0_{WY}$ and $g0_{XY} = 0_{XZ}$.

Proof: Let $h : X \rightarrow 0$, $k : 0 \rightarrow Y$. Then $g0_{XY} = gkh$. Note $gk \in \text{Hom}(0, Z)$, so $g0_{XY} \in \text{Hom}(X, Z)$ is the zero map.

Let X be an ordered set. Describe X as a category.

Answer: Let $\mathcal{C}$ have the objects X . For $x_1, x_2 \in X$, let $\mathcal{C}(x_1, x_2) = \{(x_1, x_2)\}$ if $x_1 \leq x_2$ and $\mathcal{C}(x_1, x_2) = \emptyset$ otherwise. Define the composition law by $(x_2, x_3)(x_1, x_2) = (x_1, x_3)$. Note that

$$\begin{aligned} [(x_3, x_4)(x_2, x_3)](x_1, x_2) &= (x_2, x_4)(x_1, x_2) = (x_1, x_4) \\ (x_3, x_4)[(x_2, x_3)(x_1, x_2)] &= (x_3, x_4)(x_1, x_3) = (x_1, x_4) \\ (x_2, x_2)(x_1, x_2) &= (x_1, x_2) \\ (x_1, x_2)(x_1, x_1) &= (x_1, x_2) \end{aligned}$$

Suppose $\mathcal{C}(x_1, x_2) = \mathcal{C}(x'_1, x'_2) \neq \emptyset$. Then $\{(x_1, x_2)\} = \{(x'_1, x'_2)\}$, so $(x_1, x_2) = (x'_1, x'_2)$, so $x_1 = x'_1$ and $x_2 = x'_2$.

Suppose $\mathcal{C}_0 \leq \mathcal{C}$. Let $x_1, x_2 \in X$. If $x_1 \leq x_2$, then $\mathcal{C}_0(x_1, x_2) = \{(x_1, x_2)\} = \mathcal{C}(x_1, x_2)$. Else, $\mathcal{C}_0(x_1, x_2) = \emptyset = \mathcal{C}(x_1, x_2)$.

Let G be a group. Describe G as a category with one object.

Answer: Let $\mathcal{C} = \{x_0\}$. $\mathcal{C}(x_0, x_0) = \{g : g \in G\}$. The composition rule is given by group multiplication. Identity is satisfied since $1g = g1 = g$. Group multiplication is associative, so associativity of composition is satisfied. We also ensure that every morphism has an inverse, ie that group elements have inverses. If $\mathcal{C}_0 \leq \mathcal{C}$ is full, then $\mathcal{C}_0 = \mathcal{C}$, ie that the submonoid is not proper.

Claim: The category of groups has a generator U .

Proof: Take $U = \mathbb{Z}$. Suppose $f \neq g$, so $f(x) \neq g(x)$ for some $x \in X$. Pick $u : \mathbb{Z} \rightarrow X$ such that $u(1) = x$. Then $gu(1) = g(x) \neq f(x) = fu(1)$.

Claim: There exists a one-to-one correspondence between morphisms $\mathbb{Z} \rightarrow G$ and the elements of G .

Proof: Define $\Phi : G \rightarrow \text{Hom}(\mathbb{Z}, G)$ by $\Phi(g)(1) = g$. Suppose $\Phi(g) = \Phi(g')$. Then $\Phi(g)(1) = \Phi(g')(1) \rightarrow g = g'$, so Φ is injective. Suppose $\varphi : \mathbb{Z} \rightarrow G$ is a morphism. Then $\varphi(z) = z\varphi(1)$. Let $g = \varphi(1)$. Then $\varphi = \Phi(g)$, so Φ is surjective.

I claim that the space X_0 with one element is a generator of the category of topological spaces. Suppose $f, g : X \rightarrow Y$ such that $f \neq g$. Then $f(x) \neq g(x)$ for some $x \in X$. Take $u : X_0 \rightarrow X$ by $u(x_0) = x$ (which is continuous). Thus, $fu(x_0) = f(x) \neq g(x) = gu(x_0)$.

Describe the product category. Let $\mathcal{C}_1, \mathcal{C}_2$ be categories. Let $\mathcal{C}_1 \times \mathcal{C}_2$ have the same underling objects. Let $(\mathcal{C}_1 \times \mathcal{C}_2)((X_1, X_2), (Y_1, Y_2)) = \mathcal{C}_1(X_1, Y_1) \times \mathcal{C}_2(X_2, Y_2)$. Define the composition law by $(f_2, g_2)(f_1, g_1) = (f_2 f_1, g_2 g_1)$.

Suppose $\mathcal{C}((X_1, X_2), (Y_1, Y_2)) = \mathcal{C}((X'_1, X'_2), (Y'_1, Y'_2)) \neq \emptyset$. Then $\mathcal{C}_1(X_1, Y_1) \times \mathcal{C}_2(X_2, Y_2) = \mathcal{C}_1(X'_1, Y'_1) \times \mathcal{C}_2(X'_2, Y'_2)$. The factors of the form $\mathcal{C}_1((X_1, X_2), (Y_1, Y_2))$ are nonempty. Projecting onto the first and second components gives $\mathcal{C}_1(X_1, Y_1) = \mathcal{C}_1(X'_1, Y'_1) \neq \emptyset$ and $\mathcal{C}_2(X_2, Y_2) = \mathcal{C}_2(X'_2, Y'_2) \neq \emptyset$, so $X_1 = X'_1, Y_1 = Y'_1, X_2 = X'_2, Y_2 = Y'_2$. Thus, $(X_1, X_2) = (X'_1, X'_2)$ and $(Y_1, Y_2) = (Y'_1, Y'_2)$.

I claim that composition is associative. Indeed,

$$\begin{aligned} [(f_3, g_3)(f_2, g_2)](f_1, g_1) &= (f_3 f_2, g_3 g_2)(f_1, g_1) = (f_3 f_2 f_1, g_3 g_2 g_1) \\ (f_3, g_3)[(f_2, g_2)(f_1, g_1)] &= (f_3, g_3)(f_2 f_1, g_2 g_1) = (f_3 f_2 f_1, g_3 g_2 g_1) \end{aligned}$$

Finally, we check

$$\begin{aligned} (1_{Y_1}, 1_{Y_2})(f, g) &= (1_{Y_1} f, 1_{Y_2} g) = (f, g) \\ (f, g)(1_{X_1}, 1_{X_2}) &= (f 1_{X_1}, g 1_{X_2}) = (f, g) \end{aligned}$$

2 Functors

Treating ordered sets X, Y as categories $\mathcal{C}, \mathcal{D}$, describe a functor from $\mathcal{C}$ to $\mathcal{D}$.

Let $F : X \rightarrow Y$. I claim F is order-preserving. Let $x_1 \leq x_2$, so $(x_1, x_2) \in \mathcal{C}(x_1, x_2)$. Then $F(x_1, x_2) = (F(x_1), F(x_2)) \in \mathcal{D}(F(x_1), F(x_2))$, so $F(x_1) \leq F(x_2)$. Conversely, I claim that order-preserving maps are functors. We have

$$\begin{aligned} F(x, x) &= (F(x), F(x)) \\ F((x_2, x_3)(x_1, x_2)) &= F(x_1, x_3) = (F(x_1), F(x_3)) \\ &= (F(x_2), F(x_3))(F(x_1), F(x_2)) \end{aligned}$$

Let $F : \mathcal{C}^{opp} \rightarrow \mathcal{D}$ and $f \in \mathcal{C}(Y, X), g \in \mathcal{C}(Z, Y)$. Then $f \in \mathcal{C}^{opp}(X, Y), g \in \mathcal{C}^{opp}(Y, Z)$. We have $F(gf) = \langle F(g), F(f) \rangle = F(f)F(g)$,

$$\begin{aligned} \langle \langle h', g' \rangle, f' \rangle &= \langle g' h', f' \rangle = f' g' h' \\ \langle h', \langle g', f' \rangle \rangle &= \langle h', f' g' \rangle = f' g' h' \end{aligned}$$

For the opposite category, suppose $x_1 \leq x_2$. Then $(x_2, x_1) \in \mathcal{C}^{opp}(x_1, x_2)$. We wish for $F(x_2) \leq F(x_1)$.

Define $\phi : G \rightarrow G^{opp}$ by $\phi(x) = x^{-1}$. Then $\phi(xy) = (xy)^{-1} = y^{-1}x^{-1} = \langle \phi(x), \phi(y) \rangle$, so ϕ is a homomorphism. The map $\psi : G^{opp} \rightarrow G$ defined the same way is the inverse, so $G \cong G^{opp}$.

Let $F_1 : \mathcal{C}_1 \rightarrow \mathcal{C}_2, F_2 : \mathcal{C}_2 \rightarrow \mathcal{C}_3$. Suppose $f \in \mathcal{C}_1(X_1, X_2), g \in \mathcal{C}_1(X_2, X_3)$, so (f, g) is compatible. Then $F_2 F_1(gf) = F_2(F_1(g)F_1(f)) = F_2 F_1(g)F_2 F_1(f)$ and $F_2 F_1(1_X) = F_2(1_{F_1(X)}) = 1_{F_2 F_1(X)}$, so $F_2 F_1$ is indeed a functor.

Let $F_1 : \mathcal{C}_1^{opp} \rightarrow \mathcal{C}_2$ and $F_2 : \mathcal{C}_2^{opp} \rightarrow \mathcal{C}_3$ be covariant functors. Then $F_2 F_1 : \mathcal{C}_1 \rightarrow \mathcal{C}_3$ is a functor. Let $f \in \mathcal{C}_1(X_1, X_2), g \in \mathcal{C}_1(X_2, X_3)$, so (f, g) is compatible. Then $F_2 F_1(gf) = F_2 F_1(\langle f, g \rangle) = F_2(F_1(f)F_1(g)) = F_2(\langle F_1(g), F_1(f) \rangle) = F_2 F_1(g)F_2 F_1(f)$.

Take $\mathcal{C}$ to be the category of groups and homomorphisms. Consider the map $Z : \mathcal{C} \rightarrow \mathcal{C}$ given by $G \mapsto Z(G)$. I claim that Z is not a functor. Let $f : G \rightarrow H$ be a homomorphism. Then $Z(f) : Z(G) \rightarrow Z(H)$ need not restrict to $Z(H)$. Take $G = \mathbb{Z}_2$ and $H = S_3$. We have $\mathbb{Z}_2 \hookrightarrow S_3$ since $(12) \in S_3$. We have $Z(G) = \mathbb{Z}_2$ and $Z(H) = 1$.

Consider the subcategory $\mathcal{C}'$ where morphisms are surjections (the composition of two surjections is a surjection and $1_X \in \mathcal{C}'$). I claim we can restrict $Z(f) : Z(G) \rightarrow Z(H)$. Suppose $z \in Z(G)$ and $h \in H$. Pick $g \in G$ such that $f(g) = h$. Then $f(z)h = f(z)f(g) = f(zg) = f(gz) = f(g)f(z) = hf(z)$, so $f(z) \in Z(H)$. $Z(gf) = Z(g)Z(f)$, so $Z : \mathcal{C}' \rightarrow \mathcal{C}'$ is a functor.

$Z : \text{Grp}_{\text{epi}} \rightarrow \text{Grp}$ does not restrict to $Z : \text{Grp}_{\text{epi}} \rightarrow \text{Grp}_{\text{epi}}$. Take $f : S_3 \rightarrow \mathbb{Z}_2 = S_3 / \langle (123) \rangle$. We have $Z(S_3) = 1$ while $Z(\mathbb{Z}_2) = \mathbb{Z}_2$.

Let $\Phi : \text{Ring}_{\text{comm}} \rightarrow \text{Set}$ associate to each $R \in \text{Ring}_{\text{comm}}$ the set of prime ideals of R . Then Φ is a contravariant functor.

Proof: Denote by $\text{Spec}(R)$ the set of prime ideals of R . Let $f : R \rightarrow S$ be a ring homomorphism. Then we have a map $\Phi(f) : \text{Spec}(S) \rightarrow \text{Spec}(R)$. Suppose $P \triangleleft S$ is prime (so $1_S \notin P$). I claim $f^{-1}P \triangleleft R$ is prime. First, let $r \in R$ and $p' \in f^{-1}P$. Then $f(rp') = rf(p') \in rP \leq P$, so $rp' \in f^{-1}P$. Suppose $p'_1, p'_2 \in f^{-1}P$. Then $f(p'_1 - p'_2) = f(p'_1) - f(p'_2) \in P$, so $f^{-1}P \trianglelefteq R$. Suppose $ab \in f^{-1}P$. Then $f(a)f(b) \in P$. Since P is prime, either $f(a) \in P$ or $f(b) \in P$. If $f(a) \in P$, then $a \in f^{-1}P$. Similarly for b . Finally, since $1_S \notin P$, we have $1_R \notin f^{-1}P$. Thus, $f^{-1}P$ is prime, so Φ is well-defined.

Let $f : R \rightarrow S, g : S \rightarrow T$. I claim $\Phi(gf) = \Phi(f)\Phi(g)$. Let $P \triangleleft T$ be prime. Then $\Phi(gf)(P) = (gf)^{-1}P = f^{-1}g^{-1}P = f^{-1}(\Phi(g)(P)) = \Phi(f)\Phi(g)(P)$, so $\Phi(gf) = \Phi(f)\Phi(g)$. Clearly, $\Phi(1_R) = 1_{\text{Spec}(R)}$.

For $J \trianglelefteq R$, let $V_J = \{P \in \text{Spec}(R) : P \supseteq J\}$. Let $\text{Spec}(R)$ have the closed sets V_J . We have $V_R = \emptyset$ and $V_0 = \text{Spec}(R)$. Suppose $J_1, J_2 \trianglelefteq R$. I claim $V_{J_1} \cup V_{J_2} = V_{J_1 \cap J_2}$. We have $V_{J_1}, V_{J_2} \leq V_{J_1 \cap J_2}$, so $V_{J_1} \cup V_{J_2} \leq V_{J_1 \cap J_2}$. Suppose $P \in V_{J_1 \cap J_2}$, so $P \supseteq J_1 \cap J_2$. Suppose FTSOC $P \not\supseteq J_1$ and $P \not\supseteq J_2$, so pick $a \in J_1 - P$ and $b \in J_2 - P$. Note that $ab \in J_1 J_2 \leq J_1 \cap J_2 \leq P$. WLOG $a \in P$, absurd. Now I claim $\bigcap_{\alpha} V_{J_{\alpha}} = V_{\{J_{\alpha}\}}$. Indeed, $V_{\{J_{\alpha}\}} \leq V_{J_{\alpha}}$, so

$V_{\langle \{J_\alpha\} \rangle} \leq \bigcap_\alpha V_{J_\alpha}$. Conversely, suppose $P \in \bigcap_\alpha V_{J_\alpha}$, so $P \geq J_\alpha$ for all α . Then $P \geq \langle \{J_\alpha\} \rangle$, ie $P \in V_{\langle \{J_\alpha\} \rangle}$, as desired. Thus, the V_J form a valid topology on $\text{Spec}(R)$.

Let $F : \mathcal{C}_1 \times \mathcal{C}_2 \rightarrow \mathcal{D}$ be a functor. Show that F_{C_1}, F_{C_2} are functors.

Proof: Let $f \in \mathcal{C}_1(X_1, X_2), g \in \mathcal{C}_1(X_2, X_3)$. Then

$$\begin{aligned} F_{C_2}(gf) &= F(gf, 1_{C_2}) = F(g, 1_{C_2})F(f, 1_{C_2}) = F_{C_2}(g)F_{C_2}(f) \\ F_{C_2}(1_{C_1}) &= F(1_{C_1}, 1_{C_2}) = F(1_{(C_1, C_2)}) = 1_{F(C_1, C_2)} = 1_{F_{C_2}(C_1)} \end{aligned}$$

so F_{C_2} is indeed a functor. The other part is similar.

Suppose $\varphi_1 : C_1 \rightarrow C'_1$ and $\varphi_2 : C_2 \rightarrow C'_2$. Then the diagram

$$\begin{array}{ccc} F(C_1, C_2) & \xrightarrow{F_{C_2}(\varphi_1)} & F(C'_1, C_2) \\ \downarrow F_{C_1}(\varphi_2) & & \downarrow F_{C'_1}(\varphi_2) \\ F(C_1, C'_2) & \xrightarrow{F_{C'_2}(\varphi_1)} & F(C'_1, C'_2) \end{array}$$

commutes.

Proof:

$$\begin{aligned} F_{C'_1}(\varphi_2)F_{C_2}(\varphi_1) &= F(1_{C'_1}, \varphi_2)F(\varphi_1, 1_{C_2}) = F(\varphi_1, \varphi_2) \\ F_{C'_2}(\varphi_1)F_{C_1}(\varphi_2) &= F(\varphi_1, 1_{C'_2})F(1_{C_1}, \varphi_2) = F(\varphi_1, \varphi_2) \end{aligned}$$

so the square commutes. In particular, the diagonal is $F(\varphi_1, \varphi_2)$.

Conversely, suppose we have functors $F_{C_1} : \mathcal{C}_2 \rightarrow \mathcal{D}, F_{C_2} : \mathcal{C}_1 \rightarrow \mathcal{D}$ indexed by objects of $\mathcal{C}_1, \mathcal{C}_2$ such that $F_{C_1}(C_2) = F_{C_2}(C_1)$ and the square above commutes. Then these families determine a bifunctor $G : \mathcal{C}_1 \times \mathcal{C}_2 \rightarrow \mathcal{D}$ such that $G_{C_2} = F_{C_2}$ and $G_{C_1} = F_{C_1}$.

Proof: Define $F(C_1, C_2) = F_{C_2}(C_1) = F_{C_1}(C_2)$. Suppose G has the given properties. Then $G(\varphi_1, \varphi_2) = G(1_{C'_1}, \varphi_2)G(\varphi_1, 1_{C_2}) = F_{C'_1}(\varphi_2)F_{C_2}(\varphi_1) = F_{C'_2}(\varphi_1)F_{C_1}(\varphi_2)$. Note that $G(C_1, C_2) = (C'_1, C'_2)$. We check

$$\begin{aligned} G(\varphi'_1\varphi_1, \varphi'_2\varphi_2) &= F_{C'_1}(\varphi'_2\varphi_2)F_{C_2}(\varphi'_1\varphi_1) = F_{C'_1}(\varphi'_2)F_{C'_1}(\varphi_2)F_{C_2}(\varphi'_1)F_{C_2}(\varphi_1) \\ &= F_{C'_1}(\varphi'_2)F_{C'_2}(\varphi'_1)F_{C'_1}(\varphi_2)F_{C_2}(\varphi_1) = G(\varphi'_1, \varphi'_2)G(\varphi_1, \varphi_2) \\ G(1_{C_1}, 1_{C_2}) &= F_{C_1}(1_{C_2})F_{C_2}(1_{C_1}) = 1_{F_{C_1}(C_2)}1_{F_{C_2}(C_1)} = 1_{F(C_1, C_2)} \end{aligned}$$

Show that $F : \mathcal{C}^{opp} \times \mathcal{C} \rightarrow \text{Set}$ is a bifunctor.

Proof: Suppose $X \in \mathcal{C}^{opp}$ and $Y \in \mathcal{C}$. We define $F(X, Y) = \mathcal{C}(X, Y)$. Let

$f \in \mathcal{C}^{opp}(X_1, X_2)$ and $g \in \mathcal{C}(Y_1, Y_2)$. Then $F(f, g)(h) = ghf$. Note that

$$\begin{aligned} F(f_2, g_2)F(f_1, g_1)(h) &= F(f_2, g_2)(g_1hf_1) = g_2g_1hf_1f_2 \\ &= F(f_1f_2, g_2g_1)(h) = F(\langle f_2, f_1 \rangle, g_2g_1)(h) \\ F(1_X, 1_Y)(h) &= 1_Yh1_X = h \end{aligned}$$

so F is a functor.

$g \in \mathcal{C}(Y, Z)$ is monomorphic if for all $f, f' \in \mathcal{C}(X, Y)$ that $gf = gf'$, then $f = f'$. Suppose $g \in \mathcal{C}^{opp}(Y, Z)$ is monomorphic. For all $f, f' \in \mathcal{C}^{opp}(X, Y)$, we have $\langle g, f' \rangle = \langle g, f \rangle \rightarrow f' = f$. In other words, for $g \in \mathcal{C}(Z, Y)$, $f, f' \in \mathcal{C}(Y, X)$ such that $f'g = fg$, then $f' = f$.

Show $j : Z \rightarrow \mathbb{Q}$ is an epimorphism in the category of integral domains.

Proof: Suppose $f, f' : \mathbb{Q} \rightarrow R$ such that $fj = f'j$, so $f(1) = f'(1)$. Let $m/n \in \mathbb{Q}$. We have

$$\begin{aligned} mf(1) &= mf'(1) \rightarrow f(m) = f'(m) \rightarrow nf(m/n) = nf'(m/n) \\ &\rightarrow f(m/n) = f'(m/n) \end{aligned}$$

so $f = f'$.

Show that epi, mono imply surjective, injective in Set, Top, Grp.

Proof: Suppose $f : X \rightarrow Y, g, g' : Y \rightarrow Z$ so $gf = g'f \rightarrow g = g'$. Suppose f is not surjective. Take $g : Y \rightarrow \{0, 1\}$ be constant at 0 and $g' : Y \rightarrow \{0, 1\}$ to be $g'(y) = 0$ if $y \in \text{im } f$ and $g'(y) = 1$ otherwise. Then $gf = g'f$ but $g \neq g'$.

Suppose $g : Y \rightarrow Z, f, f' : X \rightarrow Y$ such that $gf = gf' \rightarrow f = f'$. Suppose g is not injective; assume $g(y) = g(y')$ for $y \neq y'$. Take $f, f' : X \rightarrow Y$ to be constant at y, y' . Then $gf = gf'$ but $f \neq f'$.

Epimorphic implies surjective in Top. Suppose $f : X \rightarrow Y$ is not surjective. Let $g, g' : Y \rightarrow \{0, 1\}$, where $\{0, 1\}$ has the trivial topology be such that $g = 1$ and g' is the characteristic function of $\text{im } f$. Then $gf = g'f$ but $g \neq g'$, absurd. The functions above show that monomorphic implies injective in Top.

Epimorphic implies surjective in Grp. Let $\varphi : G \rightarrow H$ be epimorphic. Let K be the amalgamated product of two copies of H over $\text{im } \varphi$. Denote by θ_1, θ_2 the natural embeddings of H into K . We have $\theta_1\varphi = \theta_2\varphi$, so $\theta_1 = \theta_2$, so $\text{im } \varphi = H$, ie φ is surjective.

Monomorphic implies injective in Grp. Suppose $\varphi : G \rightarrow H$ is not injective. Pick $g_1 \neq g_2$ such that $\varphi g_1 = \varphi g_2$. Let $\psi_i : \mathbb{Z} \rightarrow G$ by $\psi_i(1) = g_i$, so $\varphi\psi_1 = \varphi\psi_2$.

3 Natural Transformations

Let $F, G : \mathcal{C} \rightarrow \mathcal{D}$. Then a natural transformation is a rule assigning each X a morphism $t_X : FX \rightarrow FY$ in $\mathcal{D}$ such that the following diagram commutes for all $f \in \mathcal{C}$:

$$\begin{array}{ccc} FX & \xrightarrow{t_X} & GX \\ \downarrow Ff & & \downarrow Gf \\ FY & \xrightarrow{t_Y} & GY \end{array}$$

If t_X is an isomorphism, then $F \cong G$ is said to be a natural equivalence. Suppose t_X is an isomorphism and the above square commutes. Then $(Gf)t_X = t_Y(Ff) \rightarrow t_Y^{-1}(Gf) = (Ff)t_X^{-1}$, so

$$\begin{array}{ccc} GX & \xrightarrow{t_X^{-1}} & FX \\ \downarrow Gf & & \downarrow Ff \\ GY & \xrightarrow{t_Y^{-1}} & FY \end{array}$$

If $t : F \rightarrow G, u : G \rightarrow H$ are natural transformations, so is $ut : F \rightarrow H$:

$$\begin{array}{ccccc} FX & \xrightarrow{t_X} & GX & \xrightarrow{u_X} & HX \\ \downarrow Ff & & \downarrow Gf & & \downarrow Hf \\ FY & \xrightarrow{t_Y} & GY & \xrightarrow{u_Y} & HY \end{array}$$

The linear map $\varphi : V \rightarrow V^{**}$ is defined by $\varphi(v)(\varepsilon) = \varepsilon(v)$ for all $\varepsilon \in V^*$ and $v \in V$. Let $f : V \rightarrow W$ be linear. We wish to show

$$\begin{array}{ccc} V & \longrightarrow & V^{**} \\ \downarrow f & & \downarrow f^{**} \\ W & \longrightarrow & W^{**} \end{array}$$

Claim: Let $F : \mathcal{C} \rightarrow \mathcal{D}$. There exists a set isomorphism between $[\mathcal{C}(A, -), F]$ and $F(A)$ given by $\tau \mapsto \tau_A(1_A)$.

Proof: Suppose $\tau_1, \tau_2 : \mathcal{C}(A, -) \rightarrow F$ are natural transformations such that $\tau_{1A}(1_A) = \tau_{2A}(1_A)$. Let $f : A \rightarrow B$. We wish to show $\tau_{1B} = \tau_{2B} : \mathcal{C}(A, B) \rightarrow F$. Consider the diagram:

$$\begin{array}{ccc} \mathcal{C}(A, A) & \xrightarrow{\tau_A} & F(A) \\ \downarrow f_* & & \downarrow Ff \\ \mathcal{C}(A, B) & \xrightarrow{\tau_B} & F(B) \end{array}$$

Then $(Ff)\tau_{1A}(1_A) = \tau_{1B}f_*(1_A) = \tau_{1B}(f)$, so $\tau_{1B}(f) = \tau_{2B}(f)$, so $\tau_{1B} = \tau_{2B}$.

Now suppose $\tau_A(1_A) \in F(A)$. Define $\tau_B(f) = (Ff)\tau_A(1_A)$. We wish to show that the following diagram commutes for $g : B \rightarrow B'$:

$$\begin{array}{ccc} \mathcal{C}(A, B) & \xrightarrow{\tau_B} & F(B) \\ \downarrow g_* & & \downarrow Fg \\ \mathcal{C}(A, B') & \xrightarrow{\tau_{B'}} & F(B') \end{array}$$

Indeed, $(Fg)\tau_B(f) = (Fg)(Ff)\tau_A(1_A) = F(gf)\tau_A(1_A) = \tau_{B'}(gf) = \tau_{B'}g_*(f)$.

Claim: $[\mathcal{C}(A, -), \mathcal{C}(A', -)]$ is in bijection with $\mathcal{C}(A', A)$, with the reverse bijection given by $\psi \mapsto \psi^*$, where $\psi : A' \rightarrow A$. Note $\psi^* : \mathcal{C}(A, -) \rightarrow \mathcal{C}(A', -)$.

Suppose $\tau \in [\mathcal{C}(A, -), \mathcal{C}(A', -)]$. Then $\tau \mapsto \tau_A(1_A)$. Note $\psi_A^* : \mathcal{C}(A, A) \rightarrow \mathcal{C}(A', A)$, so $\psi^* \mapsto \psi^*(1_A) = 1_A\psi = \psi$, so $\psi = \tau$. We have

$$\begin{array}{ccc} \mathcal{C}(A, B) & \xrightarrow{\psi_B^*} & \mathcal{C}(A', B) \\ \downarrow g_* & & \downarrow g_* \\ \mathcal{C}(A, B') & \xrightarrow{\psi_{B'}^*} & \mathcal{C}(A', B') \end{array}$$

If $\tau : \mathcal{C}(A, -) \rightarrow \mathcal{C}(A', -)$, $\tau' : \mathcal{C}(A', -) \rightarrow \mathcal{C}(A'', -)$, $\tau \mapsto \varphi$, $\tau' \mapsto \varphi'$, then $\tau'\tau \mapsto \varphi\varphi'$.

Proof: $\tau'\tau \mapsto (\tau'\tau)_A(1_A) = \tau'_A\tau_A(1_A) = \tau'_A(\varphi) = \tau'_A(\varphi_*1_A) = \varphi_*(\tau'_A(1_A)) = \varphi_*(\varphi') = \varphi\varphi'$.

Claim: $\mathbb{Z}_m \otimes \mathbb{Z}_n \cong \mathbb{Z}_{\gcd(m,n)}$.

Proof: Let $d = \gcd(m, n)$. First, I claim that the map $\tilde{f} : \mathbb{Z}_m \times \mathbb{Z}_n \rightarrow \mathbb{Z}_d$ given by $\tilde{f}(\bar{x}, \bar{y}) = \overline{xy}$ is well-defined. Indeed, let $x' = x + ma$ and $y' = y + nb$ for some $a, b \in \mathbb{Z}$. Then $x'y' = xy + nbx + may + mnab$, so $d \mid x'y' - xy$. Thus, we have the induced map $f : \mathbb{Z}_m \otimes \mathbb{Z}_n \rightarrow \mathbb{Z}_{mn}$ such that $\bar{1} \otimes \bar{1} \mapsto \bar{1}$. Conversely, define $g : \mathbb{Z}_d \rightarrow \mathbb{Z}_m \otimes \mathbb{Z}_n$ by $g(\bar{1}) = \bar{1} \otimes \bar{1}$. There exist $x, y \in \mathbb{Z}$ such that $d = mx + ny$, so

$$d(\bar{1} \otimes \bar{1}) = (mx + ny)(\bar{1} \otimes \bar{1}) = x(\overline{m} \otimes \bar{1}) + y(\bar{1} \otimes \overline{n}) = 0,$$

so g is well-defined. Note that $\mathbb{Z}_m \otimes \mathbb{Z}_n$ is cyclic with generator $\bar{1} \otimes \bar{1}$ and the generators are mapped to each other. Thus, f, g are isomorphisms and the result follows.

Claim: Let X be a spherical complex with $H_q(X; F) = 0$ for $F = \mathbb{Q}$ and $F = \mathbb{F}_p$ for all primes p and $q \geq 1$. Then $H_q(X; \mathbb{Z}) = 0$ for all $q \geq 1$.

Proof: Let $q \geq 1$ and write $H_q(X; \mathbb{Z}) = \mathbb{Z}^r \oplus \bigoplus_{p \in \mathcal{P}} A_p$, where A_p is the p -torsion of $H_q(X; \mathbb{Z})$. We have the split exact sequence

$$0 \rightarrow \text{Ext}_{\mathbb{Z}}(H_{q-1}(X; \mathbb{Z}), \mathbb{Z}_p) \rightarrow H^q(X; \mathbb{Z}_p; \mathbb{Z}) \rightarrow \text{Hom}_{\mathbb{Z}}(H_q(X; \mathbb{Z}), \mathbb{Z}_p) \rightarrow 0$$

$\text{Hom}_{\mathbb{Z}}(H_q(X; \mathbb{Z}), \mathbb{Z}_p) = \mathbb{Z}^{\text{rank}_p(A_p)}$. Note that $H^q(X; \mathbb{Z}_p; \mathbb{Z}) = H^q(X; \mathbb{F}_p)$ since group homomorphisms of abelian p -groups is the same as $\mathbb{Z}_p$ module homomorphisms. Thus, $\text{rank}_p A_p = 0 \rightarrow A_p = 0$. Since $\mathbb{Q}^r = \mathbb{Q} \otimes H_q(X; \mathbb{Z}) = H_q(X; \mathbb{Q}) = 0$, we have $r = 0$. Thus, $H_q(X; \mathbb{Z}) = 0$.

Let X be Hausdorff and $A \leq X$. (X, A) is said to be a Collared pair if $A \leq X$ is closed, for all $x \notin A$, there exist open disjoint $U(x), V(A)$, and there exists a neighborhood $B(A)$ such that there is a retract $r : B \rightarrow A$.

Claim: Let Y be Hausdorff and $f : A \rightarrow Y$ be continuous. If (X, A) is Collared, then $Z = X \cup_f Y$ is Hausdorff.

Proof: Let $p : X \sqcup Y \rightarrow Z$ be the projection. Suppose $z_1, z_2 \notin Y$, so $z_1, z_2 \in X - A$. $p^{-1}p(X - A) = X - A$ is open in $X \sqcup Y$, so $X - A$ is open in Z . Thus, z_1, z_2 can be separated by open sets in Z . Next suppose $z_1 \notin Y, z_2 \in Y$, so $z_1 \in X - A$. Pick $U(z_1)$ and $V(A)$ open disjoint. I claim that $U, p(V \sqcup Y)$ are open disjoint in Z . We have $p^{-1}p(V \sqcup Y) = V \sqcup Y$ is open. $U \leq X - A$ is open. $U, V \sqcup Y$ are saturated and $U \cap (V \sqcup Y) = \emptyset$. Finally, suppose $z_1, z_2 \in Y$. Pick $V_1(y_1), V_2(y_2)$ open disjoint in Y . Let $N_1 = p(r^{-1}f^{-1}V_1 \sqcup V_1)$ and define N_2 similarly. I claim that N_1, N_2 are open disjoint. In general for $U, V \leq X, Y$, we have

$$p^{-1}p(U \sqcup V) = (f^{-1}f(U \cap A) \cup f^{-1}V \cup U) \sqcup (f(U \cap A) \cup V)$$

Take U, V to be $r^{-1}f^{-1}V, V$. Then

$$\begin{aligned} p^{-1}N_1 &= (f^{-1}ff^{-1}V \cup f^{-1}V \cup r^{-1}f^{-1}V) \sqcup (ff^{-1}V \cup V) \\ &= (r^{-1}f^{-1}V) \sqcup V, \end{aligned}$$

which is open in $X \sqcup Y$ since $B \leq X$ is open. The sets N_1, N_2 are saturated and the corresponding $r^{-1}f^{-1}V, V$ are disjoint.

Let $\bar{f} : (E^n, S^{n-1}) \rightarrow (Z, Y)$ be such that $(E^n)^\circ \rightarrow Z - Y$ homeomorphically. Then $Z \cong E^n \cup_f Y$.

Proof: Let $f : S^{n-1} \rightarrow Y$ be the restriction of $\bar{f}$ to S^{n-1} . Define $\varphi : E^n \cup_f Y \rightarrow Z$ by extending $E^n \xrightarrow{\bar{f}} Z$ and $Y \rightarrow Z$. Note that φ sends $Y \rightarrow Y$ and $(E^n)^\circ \rightarrow Z - Y$ bijectively, so φ is bijective. I claim that φ is a closed map. Let $C \leq E^n \cup_f Y$ be closed. Let $p^{-1}C = C_1 \sqcup C_2$, where $C_1 \leq X, C_2 \leq Y$ are closed. We have from the above formula

$$\begin{aligned} p^{-1}C &= p^{-1}pp^{-1}C = p^{-1}p(C_1 \sqcup C_2) \\ &= (C_1 \cup f^{-1}f(C_1 \cap A) \cup f^{-1}(C_2)) \sqcup (f(C_1 \cap A) \cup C_2), \end{aligned}$$

where $A = S^{n-1}$. Since C_1 is closed and E^n is compact, we have that $C_1 \cap A$ is compact, so $f(C_1 \cap A)$ is closed. Thus, φ is a homeomorphism.